Retained Continuation

Three Closures of the Native Binary Ring, the Two-Lap Return, and the Price of Moving a Kept Field

From the seam of the ring to a reader that writes the field it reads

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Abstract

Transformations constitute what exists, and matter is the result of transformations that carries the potential of the transformation that made it, which is how it gets home. This paper runs that sentence on the native binary ring. The seam of every even ring holds the first ring: the two words at the center are the all-ones word with one center bit cleared, read from both sides, which is the seam 10 | 01 of R2 wrapped in ones. The compiler's triad of WHERE, WHAT and ROUTE closes in three steps with its route in the register; it continues reversibly and keeps nothing. A reader that writes its carried difference back into the field it reads keeps what it does. Its step lives in word space, and from an empty field it closes on one of three carriers, fixed by how the carry meets its mirror: a copy of R1 when the carry is the null or the one (6 steps), of R2 when the native partner acts as the complement (132 steps), and of R3 when the partner and the complement disagree (43,155,322 steps). This holds at every width. The R3 closure needs two laps. At step 21,577,661 the field is back at the mold everywhere and the reader stands on the complement of its starting word, carrying the complement of its starting difference; at step 43,155,322 everything is home. The record of the first lap is the reader's sign, and the sign is the route home. The closure first appears at width 3, the Fano width. Moving a closure means moving its kept field: the move commutes with the step, and its price is the field's gradient along the move, 13.84 cells on average for the R3 closure, 5.03 for R2 and 0.67 for the R1 sign. The R3 closure moves free only at its two vacuum ticks. A reader moved without its field leaves the field behind as a record and starts the closure its carry names. The kept field holds one write or none in every bit; unbounded occupation belongs to the nesting of widths. Every number is printed by the two shipped programs.

1. The claim and the route

Transformations constitute reality. Verbs precede nouns, and a noun is a pack of encapsulated verbs. Matter is not a substance that transformations act on. It is the result of binary transitions, and it carries the potential

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of the transformation that made it, which is how it gets home; matter does not move, it transitions, and the rings are the memory map of those transitions (Kulik 2026a).

The native binary ring R_n holds every n-bit word at one address. Nothing in it is built; it is read. The program's earlier work read the ring as a compiler: a triad of WHERE, WHAT and ROUTE that closes in three steps, a route switch that rebuilds the ring with no anchor, and a four-step diamond at the innermost shell that holds its place at every width (Kulik 2026b; 2026c; 2026e). That compiler continues reversibly and keeps nothing. This paper follows one question out from the center of the ring: what has to be added for a transformation to keep its own result and still get home, and what the result is when it does.

The route runs from the center out. Section 2 reads the seam of the ring and finds the first ring there. Section 3 sets out the compiler, which is reversible continuation. Section 4 lets the reader write the field it reads, which is retained continuation, and section 5 counts every closure an empty field can start. Section 6 follows the closure that needs two laps. Section 7 moves it and prices the move. Section 8 reads what the kept field holds and what a finite field cannot hold. Section 9 pulls the route into one view, and section 10 is the ledger.

Statuses follow the program's ledger. RUNS marks what the shipped code computes on every case named. STANDS marks results that proof or established theory supplies. OPEN marks a missing link that is named and not yet written. DEAD END marks code that ran and landed where the landing belongs to another owner.

2. The center of the ring

2.1 The seam holds the first ring

R_n lists the representatives, the words x with $x < P(x)$, in increasing order, then their partners in reverse, so that the facing law $R[N - 1 - a] = P(R[a])$ holds at every address (Kulik 2026c). P is the native partner: reversal J on every word that reversal moves, complement C on the palindromes. The seam lies between addresses $h - 1$ and h, with $h = 2^{n-1}$, where the last representative meets its partner.

At even width $n = 2r$ write u for the all-ones word and $e_L = 2^r$, $e_R = 2^{r-1}$ for the two center bits. The last representative is $u \oplus e_L$. Every word above it begins with r ones; either it is u, whose partner is 0, or its reversal is smaller than it, so none of them is a representative. Its own reversal $u \oplus e_R$ is larger, so it is one. The seam is therefore

$$R[h - 1] = u \oplus e_R, \quad R[h] = u \oplus e_L$$

at every even width: the one with a single center bit taken out, read from both sides. At R2 these are 10 and 01, the first ring's own seam, and at every even width above it the seam is that pair wrapped in ones (Table 1). The two words across the seam differ by exactly the center pair $s = e_L \oplus e_R$. **RUNS** from R2 to R14, and the argument above carries it to every even width.

Table 1. The seam of the even rings. Each seam word is the all-ones word with one center bit cleared; the two differ by the center pair.

Ring	R[h – 1]	R[h]	Across the seam
R2	10	01	11

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Ring	R[h − 1]	R[h]	Across the seam
R4	1101	1011	0110
R6	111011	110111	001100
R8	11110111	11101111	00011000
R10	1111101111	1111011111	0000110000
R12	111111011111	111110111111	000001100000

On odd widths the center is a single bit $c = 2^{(n-1)/2}$, a palindrome. The seam there is the one with a bit next to the center taken out (11101 | 10111 at R5), and c sits at address c, in the opening run of the ring where every word reads as its own address. So the question the center raises has a direct answer. The last spot at the center of every even ring is the first ring's difference: a single bit taken out of the one, read from both sides.

2.2 The center carrier

The words u, e_L and e_R span an eight-word space L. Its words are the first ring on the center pair, 0, e_L, e_R and s (00, 10, 01 and 11 on the two center bits, zeros outside), and their complements. L is closed under facing, and its eight words sit at the ring's first address (0), its last (u), its seam ($u \oplus e_R$ and $u \oplus e_L$) and on two facing chords: e_R at 2^{r-1} facing e_L, and s at $3 \cdot 2^{r-1} - 1$ facing $u \oplus s$ (Figure 1). The center carrier holds the start, the end and the seam of every even ring. **RUNS** from R4 to R14.

The innermost diamond of the running audit, $1^d00011^d \rightarrow 1^d00111^d \rightarrow 1^d01111^d \rightarrow 1^d01011^d$ with d = r − 2, sits at seam offsets [h − 6, h − 4, h − 2, h − 3] at every even width from R6 (**RUNS** through R128 in the earlier audits; Kulik 2026e, §239). Its four words are $m_* \oplus \{0, e_R, s, e_L\}$ with $m_* = 1^d00011^d$: the first ring again, placed at m_*, stepping by the two center bits. The diamond's carrier $m_* \oplus L$ is the center carrier at another placement. A nesting that cannot nest any more is the first ring.

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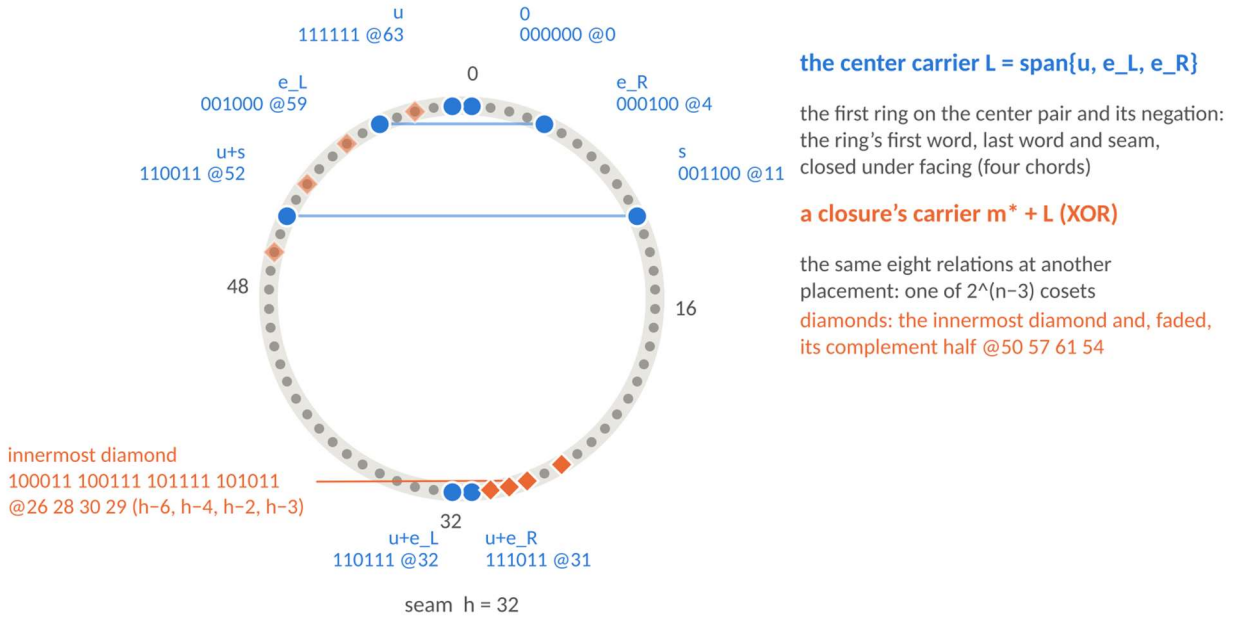


Figure 1. The center carrier on R6. The 64 addresses run clockwise from 0 at the top; the seam lies between 31 and 32. Blue: the eight words of L = span{u, e_L, e_R} with their addresses, joined by their facing chords. They hold the ring's first word, its last word and its seam. Orange: the innermost diamond at h − 6, h − 4, h − 2 and h − 3, and, faded, the complement half of its carrier $m_* \oplus L$.

3. The compiler: reversible continuation

3.1 One byte

R2 read once round is the stored circle 00 10 01 11, the byte 0x27. Its cyclic difference $(I \oplus T)\cdot 0x27$ is 0x69, the truth table of the three-input sum, and the three-input carry 0x17 is read from the same circle (Kulik 2026e, §§203–204). Reversing the byte is an address jump, $J_8(0x27) = T^5(0x27)$. The sheet reading $C_8J_8(0x27)$ = 0x1B has the difference 0x2D, which is the diamond's four states 00 → 10 → 11 → 01 written in one byte. The three-point switch $(I \oplus T \oplus T^2)\cdot 0x27$ = 0xF5 rebuilds 0x27 with no anchor bit. One distinction read round a circle already holds the adder, the diamond and its own inverse. **RUNS.**

3.2 The triad

A read at address a of value v binds as Γ(a, v) = (READ(v), v ⊕ R[a]). Call m = R[a] the WHERE, v the WHAT and d = m ⊕ v the ROUTE. Then m ⊕ v ⊕ d = 0 and

$$\Gamma(m, v, d) = (v, d, m), \quad \Gamma^3 = I, \quad \Gamma^2 = \Gamma^{-1}.$$

The roles rotate, and compiling twice is unbinding. The route home is part of the state, so the compiler runs with no stack. The 4^n states split as $1 + 3\cdot(4^n - 1)/3$: the origin, $2^n - 1$ zero-gap triads, and $(2^n - 1)(2^n - 2)/3$ interior triads, which are the oriented lines of the projective space PG(n − 1, 2). At n = 3 these are the seven lines of the Fano plane (Fano 1892). Through any nonzero WHERE pass $2^{n-1} - 1$ lines. At n = 3 three loops of

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three meet at each point and cover the space through shared ownership, $3 \times 3 - 2 + 1 = 8$. **RUNS** for $\Gamma^3 = I$ on R1 to R6 and for the counts.

3.3 The route switch

Lift each ring edge to a triad $E(a) = (x_a, x_{a+1}, x_a \oplus x_{a+1})$. The compiler predicts $\Gamma E(a)$, and the ring's next edge is $E(a + 1) = Q_t(\Gamma E(a))$ with $Q_t(m, v, d) = (m, v \oplus t, d \oplus t)$ and $t = t_a = x_a \oplus x_{a+1} \oplus x_{a+2}$. Over the whole ring $t = (I \oplus T \oplus T^2)X$. The two-point difference $I \oplus T$ is nilpotent. The three-point switch is invertible, $(I \oplus T \oplus T^2)^N = I$ for $N = 2^n$, so $X = Q_{op}{}^{N-1}t$ rebuilds the ring with no anchor bit. The switch vanishes where the ring agrees with the compiler: at $a = 1$ on R2, $a = 3$ on R3, $a \in \{1, 3, 12\}$ on R4, and at $a = 1$ and $a = 2^n - 4$ on every ring from R5. The perimeter edges visit 5 lines on R3, 11 on R4 and $2^n - 4$ from R5 on. On R3 the two remaining Fano lines close through the header flip and through the palindrome chord {010, 101, 111}, the line no perimeter edge can walk. **RUNS** from R2 to R10.

3.4 The diamond

A carried value v at a fixed mold $M = R_n$ returns as $U_B(a, v) = (\text{READ}(v), v \oplus B(v \oplus M[a]))$. In the frame (m, Δ) this is $(m, \Delta) \mapsto (m \oplus \Delta, B(\Delta))$, and with $B = P$, which fixes no word, every orbit closes in four steps. The innermost diamond is this orbit with $\Delta = e_R$. On the residue pair square $W_r(i, j) = i \parallel J_r(j)$ it is the 2×2 cell of two rungs and two register slips, with $\text{gap}_{S-1} - \text{gap}_{S+1} \in \{0, 1\}$ on every off-diagonal cell (Kulik 2026c, §6).

Every orbit in this section returns because the mold never changes. A closure leaves no trace in the ring it ran on. Reversible continuation is the way home; it is not yet anything that stays.

4. The field: retained continuation

4.1 The step

Let the reader write. Hold the mold $M = R_n$ fixed and give the ring a live value field V, starting at $V = M$. A reader at address a carrying the difference w reads the effective value $e = V[a] \oplus w$, moves to $b = \text{READ}(e)$, carries $w' = P(e \oplus M[a])$, and writes $V'[a] = V[a] \oplus w'$. No other cell changes. The step undoes exactly:

$$a = \text{READ}(M[b] \oplus P(w')), \quad V[a] = V'[a] \oplus w', \quad w = M[b] \oplus V[a].$$

The step was set to the program in the audits of 1 November, which found it confined to eight addresses with a full return after 43,155,322 steps (running audit; ChatGPT session).

READ undoes the mold, so the step lives in word space. Write $p = M[a]$ for the reader's word and $D = V \oplus M$ for the field's offsets:

$$x = D[p] \oplus w, \quad p' = p \oplus x, \quad w' = P(x), \quad D'[p] = D[p] \oplus w'.$$

No address order enters. The reader moves by XOR, which is how address moves on the ring (Kulik 2026e, §224), and the ring says where each word sits. Matter does not move, it transitions, and the ring is the map of where.

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4.2 Carriers

A linear space K that contains u and is closed under reversal is closed under P, because P is reversal or complement and both keep K. A reader born on an empty field with carry w therefore never leaves p ⊕ K(w), with K(w) = span{u, w, J(w)}, and nothing it writes leaves that set either. **STANDS.**

K(w) has one of three sizes, fixed by how the carry meets its mirror. If w is the null or the one, K(w) = {0, u}, a copy of R1. If w is a palindrome, or an anti-palindrome whose reversal is its complement, then J(w) is w or w ⊕ u and K(w) holds four words, on every one of which P is the complement: a copy of R2. Otherwise w, J(w) and u are independent and K(w) holds eight words, a copy of R3, on which P is reversal on some words and complement on others. In the coordinates (u, w, J(w)) the step on an eight-word carrier is the same at every width. Every reader an empty field can start carries one of the first three rings with it, at every width, and the rest of the ring is placement.

5. Three closures

Table 2 runs every carry on an empty field: in the full word space at widths 2, 3, 4 and 5, every carry; on the eight-word carrier in coordinates, which holds at every even width from 4; and on the full rings R6 to R12 in the audits of 2 November. Three periods appear, at every width: 6, 132 and 43,155,322.

Table 2. Every closure an empty field can start. The class is fixed by how the carry meets its mirror. Mean kept cells are averages over each closure's own period.

Carrier	Carries, even n	Carries, odd n	Period	Ticks with the field at the mold	Mean kept cells	At the half lap
R1: {0, u}	2	2	6	4 of 6	0.33 of 2	reader on the complement word
R2: P = C	$2^{n/2+1} - 2$	$2^{(n+1)/2} - 2$	132	1	2.52 of 4	no half lap
R3: P ≠ C	$2^n - 2^{n/2+1}$	$2^n - 2^{(n+1)/2}$	43,155,322	2	6.92 of 8	field at the mold, reader complemented

The count of R3 carries is the count of words on which the native partner and the complement disagree, the discrepancy law of the record. R2 has no such word. On the first ring the mirror pair 01 | 10 is also a complement pair, so every carry on R2 closes in 6 or 132 steps and none needs two laps. The two-lap closure first appears at width 3, the Fano width, the first width at which a word and its mirror can differ by less than the whole word.

In verbs: where reflecting and negating are one transformation (P = C), the reader only negates, and it closes as the sign alone (R1) or in one lap (R2). Where reflecting and negating are two transformations (P ≠ C), the reader interleaves them, keeps a field, and needs two laps (R3).

With the mirror alone, J in place of P, the same carries close in 1, 6 and 518 steps. The R3 carries still need two laps, but at the half lap the reader comes back mirrored, e_R turned into e_L, not complemented. The

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palindrome branch of the native partner turns a half-lap mirror into a half-lap sign, and it stretches the clock from 518 = 2·7·37 to 43,155,322 = 2·7·3,082,523. **RUNS.**

6. Two laps home

Seed the R3 closure on the innermost diamond: p = m_*, w = e_R, D = 0. Its first four steps are the diamond. On the fifth the reader reaches m_* again and finds its own first write there. The effective difference is the palindrome s, the native partner complements it, and the direction u opens. From then on the closure fills its carrier. Over the full period the field keeps 6.92 of its 8 cells on average, keeps 7 or 8 on 70.6 percent of ticks, and is back at the mold on exactly two ticks (Table 3; Figure 2a).

Table 3. The kept field over one full period of the R3 closure, 43,155,322 ticks. Mean 6.919906.

Kept cells k	0	1	2	3	4	5	6	7	8
Ticks	2	106	3,212	50,576	475,298	2,738,622	9,409,712	17,402,406	13,075,388

One of the two vacuum ticks is the end of the period. The other is its exact middle. At step 21,577,661 every field cell is back at the mold, and the reader stands at $m_* \oplus u$, the complement of its starting word, carrying $e_R \oplus u$, the complement of its starting difference. In the ×2 mod $(2^n - 1)$ reading of the ring the complement is the negative, C(x) = −x (Kulik 2026e, §203). The reader comes back as the negative of itself, with the field at +1. The field cannot tell one lap from none; the reader can. At step 43,155,322 everything is home.

The half-lap state is the image of the start under an involution σ that commutes with the step. The program finds σ among the 32 symmetries of the step on the carrier, built from the complement and the carrier's translations, and checks the commutation on 20,000 random states. The orbit is a double cover of an orbit of odd length, 21,577,661: one swap with no fixed point, two laps home, the twisted ring of Two Laps Home (Kulik 2026d). The record of the first lap is held by the reader's sign alone, and that sign is the route home. This is the abstract of the Reference Record as a computation: matter carries the potential of the transformation that made it, which is how it gets home.

Established physics holds the same path at a different location. A spin-½ state turned through 360° returns with its sign reversed and through 720° returns to itself, as neutron interferometry measures (Rauch et al. 1975; Werner et al. 1975), and Law 4 of the foundational cascade places odd-parity closures there (Kulik 2026b). **STANDS** for the comparator and **RUNS** for the closure.

The R1 closure has a half lap too. Born with carry 0, it writes u, erases it, steps to the complement word, writes u, erases it and steps back: three steps out to −p and three home, with the field written on two of its six ticks. It is the triad's three steps with a sign, the null and the one coming back flipped (Kulik 2026e, §236): the sign with nothing kept. The R2 closure keeps a field and has no half lap; it never meets its own complement and comes home in one lap of 132. The R3 closure is the one that carries the sign and keeps a field.

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7. Moving the closure

7.1 The move carries the field

A closure is placed by its carrier. The placements of an eight-word carrier are the cosets of L, 2^{n-3} of them at width n. To move a closure from one placement to another, the reader's word and the kept field move together, and the mold stays where it is:

$$H_h(p, w, D) = (p \oplus h, w, q \mapsto D[q \oplus h]).$$

H_h commutes with the step, $H_hF = FH_h$, with $H_h^2 = I$ and $H_aH_b = H_{a\oplus b}$: the step reads the offset under the reader, and a moved reader over a moved field reads the same offset (ChatGPT audit of 2 November, at four widths and on 1,250 random states). A moved closure keeps its period and its half lap; it is the same route at another WHERE. The next shell out, $o_R = 2^{r-2}$ and its mirror $o_L = 2^{r+1}$, opens two directions in the quotient, and the four carriers A, A $\oplus$ o_R, A $\oplus$ o_L and A $\oplus$ o_R $\oplus$ o_L form a square whose edges are reversible moves. The address gaps those moves span on the ring are not constant: one move takes the anchor from address 26 to 39 at R6 and from 2,042 to 3,861 at R12. The quotient gives the relational distance; lengths need a further reading.

7.2 Three handoffs

Table 4 sets four handoffs side by side, each applied at tick 4,096, when seven field cells are kept. The audits ran them at R6, R8, R10 and R12, with identical results at every width.

Table 4. Handoffs into a closure that keeps a field (ChatGPT audits of 2 November).

Handoff at tick 4,096	Words visited in the next 100,000 ticks	Where it lands
none	8	the two-lap closure
carry only: w ← w $\oplus$ o_R	32	the kick and the kept field open W = span{u, e_L, e_R, o_L, o_R}: PG(4, 2), 31 points, 155 lines
reader only: p ← p $\oplus$ o_R	2	a six-step, two-word closure; the seven kept cells stay behind
reader and field: H_h, h = o_R	8	the two-lap closure at the new placement, full period kept

The kept field owns the opening to 32 words. The same carry kick applied at birth, or at the half-lap vacuum, stays on eight words; it changes which eight-word closure forms, not how many words it reaches.

7.3 The reader alone carries its class

The six-step, two-word closure is the R1 sign: at tick 4,096 the reader's carry is u. A reader moved off its field into an empty placement finds vacuum there, and from an empty field the class of its carry fixes what it does. It starts the R1 sign, the R2 closure or a new R3 closure, and the field it left keeps its writes with no reader to change them. Over the full period the carry is in the R1 class on 23.08 percent of ticks, the R2 class on 24.22

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percent and the R3 class on 52.70 percent (Figure 2a). Those are the shares in which a reader pulled from its field becomes the sign, the one-lap closure or a new two-lap closure. **RUNS.**

So the closure has two halves with two owners. The reader carries the class and the sign; the kept field carries what a move has to carry. Taken apart, the reader is reborn as the closure its carry names and the field stays behind as a record, the ring as the memory map of the transition that wrote it.

7.4 The price of moving

The field cells that differ between a state and its moved image are the price of the move:

$$C_{cell}(D; h) = \Sigma_q [D(q) \neq D(q \oplus h)],$$

the field's gradient along h. For a move out of the carrier the old and new supports are disjoint, and with k cells kept $C_{cell} = 2k$: 14 cells at tick 4,096, with 28 to 52 field bits changed as the width runs from 6 to 12. Averaged over each closure's own period the price is 13.84 cells for the R3 closure, 5.03 for the R2 closure and 0.67 for the R1 sign (Figure 2b).

The R3 closure moves free on exactly two of its 43,155,322 ticks, the half lap and the full lap, when the whole record is in the reader's sign and the field is at the mold. There H_h is the single XOR $p \leftarrow p \oplus h$, and the moved closure runs its full period at the new placement. The R1 sign moves free on four of its six ticks.

The foundational cascade sets the terms: a transformation must pay a price to prove that something changed, the price is a gradient and not a single value, and mass must have a gradient (Kulik 2026b). The price here is that gradient, and it is a gradient of the kept field. The kept field is what a move must carry, and the sign alone carries almost nothing and moves almost free. Mass is the kept field of a closure, read as the price of moving it, and the three closures order the same way at every width: the sign, the one-lap closure, the two-lap closure. **RUNS** for the counts. What selects and executes a move from inside the step is **OPEN** (section 10).

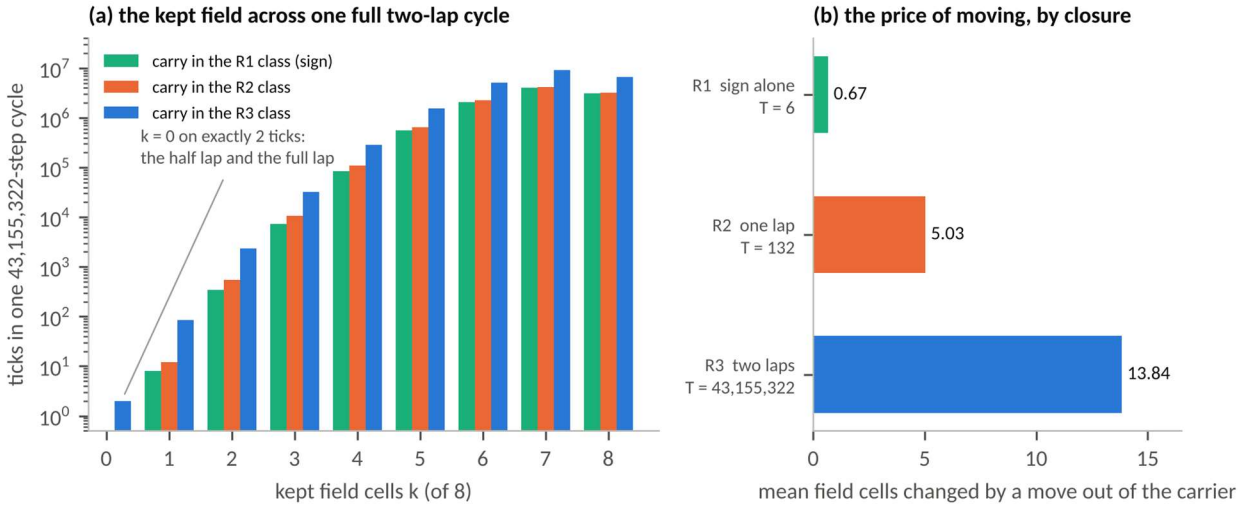


Figure 2. The kept field and the price of moving it. (a) Ticks of one full 43,155,322-tick period of the R3 closure by the number of kept field cells, split by the class of the reader's carry at that tick; log scale. The field is at the mold on exactly two ticks. (b) The mean number of field cells a move out of the carrier changes, 2k averaged over each closure's own period.

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8. What the kept field holds

8.1 One write per place

Every bit of every field cell is a place that holds a write or does not. A write toggles it, and the same difference written twice returns it. The field on an eight-word carrier has 24 such places and 2^{24} states, the state count of 24 modes that are each occupied or empty, which is the state count of a fermionic field of 24 modes (Pauli 1925; Jordan & Wigner 1928). The kept field holds occupation one write deep, and the reader that writes it needs two laps. Both halves of the matter signature sit on one carrier.

8.2 What a finite field cannot hold

A finite field cannot hold unbounded occupation. A closure on a finite carrier repeats its states after its period, so it cannot count laps without end. Two identical writes cancel, so the field holds parity, not number. The bosonic relation $[a, a^{\dagger}] = I$ has no solution in finite dimension, because the trace of a commutator is zero and the trace of the identity is not (Fock 1932; reverse-saturation audit of 2 November). Unbounded occupation needs an unbounded place, and the ring has exactly one: the widths nest without end. You cannot outrun the compiler; you can only nest. Bosonic multiplicity belongs to the nesting of widths. One-write occupation and the two-lap return belong to a single width. The law that composes several readers on one field, keeps each return, and lets readers act on each other where their carriers overlap is the next WHAT to write. **OPEN**, with its type named.

8.3 Saturation belongs to the reader

The two-point reader $I \oplus T$ cannot tell a field X from its complement: $(I \oplus T)(X \oplus 1) = (I \oplus T)X$. The three-point reader can, $Q(X \oplus 1) = Q(X) \oplus 1$, and it has an inverse. The ring is the same under both questions; the reader decides what saturates. A reader's capacity is $\log_2$ of the number of fields it can tell apart, and a reader can saturate while the field does not.

9. The route in one view

Start at the center and read out. The seam of every even ring is the first ring's difference, a single center bit taken out of the one and read from both sides. The center carrier is the first ring and its negation, and it holds the start, the end and the seam of the ring. The compiler runs the triad through it: WHERE, WHAT and ROUTE rotate and close in three steps with the route in the register, so continuation needs no stack and keeps nothing. When the reader writes the field it reads, the route moves out of the register and into the field. From an empty field the reader closes on a copy of R1, R2 or R3, chosen by how its carry meets its mirror. Where the mirror and the complement agree the reader only negates. Where they disagree it keeps a field and needs two laps, and at the half lap the whole record of the first lap is the reader's sign. That closure first appears at R3, where three loops of three cover the space. Moving it means moving its kept field. The price is the field's gradient, and the sign alone moves almost free. Pull the reader off its field and it is reborn as the closure its carry names, while the field stays behind as a record. The kept field holds one write per place, and unbounded occupation lives in the nesting of widths.

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The squares the route draws

The seam is to every ring as the first ring's 10 | 01 is to the center that every ring holds.

The compiler is to the field as reversible continuation is to retained continuation.

Γ, home in three steps, is to the R3 closure, home in two laps, as a route held in the register is to a route held in the reader's sign.

P = C is to P ≠ C as a reader that only negates is to a reader that keeps a field and needs two laps.

The reader is to the kept field as the class and the sign are to the price of moving.

Moving reader and field together is to moving the reader alone as the same closure at a new place is to a reader reborn and a record left behind.

Matter is the result of binary transitions and carries the potential of the transformation that made it, which is how it gets home. In the runs that sentence is the R3 closure. The result is the kept field, the potential is the reader's sign, and home is the second lap.

10. Ledger

Table 5. Ledger.

Status	Result
RUNS	The seam $u \oplus e_R \mid u \oplus e_L$ on R2 to R14, with the argument of section 2.1 for every even width; the addresses of the center carrier on R4 to R14.
RUNS	The compiler identities: 0x27 → 0x69, 0x1B → 0x2D, $J_8 = T^5$ on 0x27, 0xF5; $\Gamma^3 = I$ on R1 to R6 and the triad counts; the switch zeros and perimeter lines on R2 to R10.
RUNS	The word-space step and its inverse; the three closures from an empty field, every carry, full word space at widths 2 to 5, the eight-word carrier at every even width from 4, and the full rings R6 to R12.
RUNS	The half lap at step 21,577,661: field at the mold, reader complemented, image of the start under an involution of the step; the mirror-only half lap at 259 of 518.
RUNS	The kept-field distribution over the full period; the move H_h and its commutation; the full period after a move; the free move at the vacuum ticks; the four handoffs.
RUNS	The R1 class of the reader-only packet; the carry-class shares 23.08, 24.22 and 52.70 percent; the price, 14 cells at tick 4,096 and cycle means 13.84, 5.03 and 0.67.
STANDS	The facing law of the native ring; P-stability of every reversal-stable space containing u; the finite-dimensional obstruction to $[a, a^\dagger] = I$; spinor sign at 360° and identity at 720° as comparator.
DEAD END	A carry-only kick of a kept field as transport: it lands on the 32-word opening PG(4, 2), owned by the kept field.
DEAD END	A reader-only kick as transport: it lands on the R1 sign and a record left behind.

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Status	Result
DEAD END	The closing claim of the 1 November draft that the innermost diamond is a self-sustaining physical structure: the diamond is an orbit of a fixed mold and keeps nothing; keeping belongs to the field step.
OPEN	What inside the step selects h and carries the field with the reader: the move from inside.
OPEN	The acceleration law, and which factor of the constant −1 of the angle law $u = t - t^2/2$ is mass and which acceleration (record entry NESTED).
OPEN	Exchange: two readers on one field and the sign of swapping them; the composition law for several readers; bosonic occupation from the nesting of widths.
OPEN	The common factor 2·7 of 518 and 43,155,322; the metric reading of placements, since address gaps are not lengths.

Reproduction

Two programs print every number in this paper. **NEXUS_Retained_Continuation.c** (gcc -O2, about half a minute) runs every carry on an empty field in the full word space at widths 2 to 5 and the eight-word carrier in coordinates: periods, vacuum ticks, the half-lap image and its symmetry, the commutation check, the kept-field distribution by carry class, the state at tick 4,096, and the mirror-only clock.
NEXUS_Retained_Continuation_center.py (standard library, a few seconds) builds the native rings and prints the seam, the center carrier's addresses, the diamond offsets and the compiler identities. Their outputs ship beside them. The handoff and move results of section 7.1 and 7.2 are the ChatGPT audits of 2 November, with their sealed predictions and traces.

Technical note (Claude, Anthropic)

1. **The two-lap state (section 6).** The run computes an exact half-period image of an integer orbit under an involution of the step. It builds no rotation group and no state space with amplitudes; the spinor path is a comparator, and no spin magnitude follows from the period.
2. **The price (section 7.4).** C_{cell} counts field values that differ between a state and its moved image. It is not an executed cost: H_h is a symmetry of the step applied from outside, and the step does not perform it on its own. The identification of the price with mass rests on that count; an inertial law needs the move from inside, listed OPEN.
3. **Width (sections 4 and 5).** The three-closure law follows for every width from the carrier lemma and the coordinate form of the step. The full word-space runs cover widths 2 to 5, the coordinate runs every even width from 4, and the audits of 2 November the full rings R6 to R12.
4. **The native ring (sections 2 and 3).** The ring is the facing construction: representatives ascending, partners descending. It reproduces R2 = 00 10 01 11, the stored circle 0x27 and the diamond's seam offsets. Statements about addresses (the seam, the carrier's addresses, the switch zeros) are statements about this order. The closure law is not, because the step lives in word space.

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5. **One write per place (section 8.1).** 2^{24} is the count of states of 24 binary occupations. The sign under exchange that a fermionic field also carries is not shown here; it is listed OPEN.
6. **Seeds (sections 6 and 7).** The period, the half lap, the kept-field distribution and the carry shares are read on the closure seeded at the innermost diamond with carry e_R. Every R3 seed is the image of this one under a symmetry of the step and a change of carrier coordinates.
7. **Corrections to the 1 November draft (Gemini).** Its list of switch zeros dropped the second zero at $a = 2^n - 4$ from R5 on. It used the facing law and the palindrome isolation theorem without deriving them, and its closing sentence claimed more than a fixed-mold orbit carries. Section 3 restates its compiler results in corrected form.

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```c
/* Retained Continuation: reproduction of the closure results.
   Dean A. Kulik, QuHarmonics Research Group, ORCID 0009-0003-3128-8828. Runs by Claude (Anthropic), November 2026.
   Build and run:  gcc -O2 -o rc NEXUS_Retained_Continuation.c && ./rc        (about half a minute)

   The coupled reader-field step in word space (READ undoes the mold, so no address order enters):
        x = D[p] ^ w ;  p' = p ^ x ;  w' = P(x) ;  D[p] ^= w'
   P = native partner: reversal J off the palindromes, complement on the palindromes.

   Part A  every carry on an empty field, full word space, widths 2..5: period and every return of the field to the mold.
   Part B  the eight-word carrier span{u, eR, eL} in coordinates (bit0 = eR, bit1 = eL, bit2 = u), valid at every even
           width >= 4: the 32 symmetries of the step and their commutation, the half-lap image, the mirror-only clock,
           the kept-cell distribution over the full period by carry class, and the state at tick 4,096.        */
#include <stdio.h>
#include <stdint.h>
#include <stdlib.h>
#include <string.h>

/* ---------------- Part A: full word space ---------------- */
static int n, N; static uint32_t U;
static uint32_t Jw(uint32_t x) { uint32_t y = 0; for (int i = 0; i < n; i++) if (x >> i & 1) y |= 1u << (n - 1 - i); return y; }
static uint32_t Pw(uint32_t x) { uint32_t j = Jw(x); return j == x ? x ^ U : j; }
static void bits(uint32_t x, char *b) { for (int i = 0; i < n; i++) b[i] = '0' + ((x >> (n - 1 - i)) & 1); b[n] = 0; }
static const char *cls_full(uint32_t w) {
    if (w == 0 || w == U) return "R1";
    uint32_t j = Jw(w); if (j == w || j == (w ^ U)) return "R2";
    return "R3";
}
static void partA(int width) {
    n = width; N = 1 << n; U = N - 1; uint8_t D[64];
    long long per[3] = {0, 0, 0}; int count[3] = {0, 0, 0};
    for (uint32_t w0 = 0; w0 < (uint32_t)N; w0++) {
        memset(D, 0, N); uint32_t p = 0, w = w0; long long t = 0; int nv = 0; char half[64] = "";
        for (;;) {
            uint32_t x = D[p] ^ w, w2 = Pw(x); D[p] ^= w2; p ^= x; w = w2; t++;
            int vac = 1; for (int i = 0; i < N; i++) if (D[i]) { vac = 0; break; }
            if (vac) {
                nv++;
                if (p == 0 && w == w0) break;
                if (!half[0]) { char bp[16], bw[16]; bits(p, bp); bits(w, bw); snprintf(half, sizeof half, "t=%lld p=%s w=%s", t, bp, bw); }
            }
        }
        const char *c = cls_full(w0); int k = c[1] - '1';
        char bw0[16]; bits(w0, bw0);
        printf("  n=%d carry %s  %s  period %-9lld  field at the mold on %d ticks%s%s\n", n, bw0, c, t, nv, half[0] ? "  first: " : "", half);
        if (count[k] && per[k] != t) { printf("  CLASS PERIOD MISMATCH\n"); exit(1); }
        per[k] = t; count[k]++;
    }
    printf("  n=%d summary: R1 %d carries (period %lld), R2 %d carries (period %lld), R3 %d carries (period %lld)\n",
           n, count[0], per[0], count[1], per[1], count[2], per[2]);
}
```

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```
 51.
 52. /* ---------------- Part B: the eight-word carrier in coordinates ---------------- */
 53. static int USEP = 1;
 54. static inline uint32_t PART(uint32_t x) { uint32_t cR = x & 1, cL = (x >> 1) & 1; if (USEP)
return (cR == cL) ? (x ^ 4) : (x ^ 3); return (x & 4) | (cR << 1) | cL; }
 55. static inline uint64_t step(uint64_t s) {
 56.     uint32_t p = s & 7, w = (s >> 3) & 7, D = (uint32_t)(s >> 6);
 57.     uint32_t x = ((D >> (3 * p)) & 7) ^ w, w2 = PART(x); D ^= w2 << (3 * p);
 58.     return (uint64_t)(p ^ x) | ((uint64_t)w2 << 3) | ((uint64_t)D << 6);
 59. }
 60. static const char *nm(uint32_t x) { static const char *T[8] = {"0", "eR", "eL", "s", "u",
"u+eR", "u+eL", "u+s"}; return T[x & 7]; }
 61. static uint32_t app(const uint32_t *g, uint32_t x) { uint32_t y = 0; for (int i = 0; i < 3;
i++) if (x >> i & 1) y ^= g[i]; return y; }
 62. static uint64_t act(const uint32_t *g, uint32_t c, uint64_t s) {
 63.     uint32_t p = s & 7, w = (s >> 3) & 7, D[8], E[8];
 64.     for (int m = 0; m < 8; m++) D[m] = (s >> (6 + 3 * m)) & 7;
 65.     for (int m = 0; m < 8; m++) E[app(g, m) ^ c] = app(g, D[m]);
 66.     uint64_t r = (app(g, p) ^ c) | (app(g, w) << 3); for (int m = 0; m < 8; m++) r |=
(uint64_t)E[m] << (6 + 3 * m);
 67.     return r;
 68. }
 69. static int carry_class(uint32_t w) { if (w == 0 || w == 4) return 0; if (w == 3 || w == 7)
return 1; return 2; }
 70.
 71. static void partB(void) {
 72.     /* symmetries: linear maps commuting with the partner, times the 8 carrier translations */
 73.     uint32_t G[168][3]; int NG = 0;
 74.     for (uint32_t a = 1; a < 8; a++) for (uint32_t b = 1; b < 8; b++) for (uint32_t c = 1; c <
8; c++) {
 75.         if (b == a || c == a || c == b || c == (a ^ b)) continue;
 76.         uint32_t g[3] = {a, b, c}; int ok = 1;
 77.         for (uint32_t x = 0; x < 8 && ok; x++) if (app(g, PART(x)) != PART(app(g, x))) ok = 0;
 78.         if (ok) { memcpy(G[NG], g, sizeof g); NG++; }
 79.     }
 80.     srand(7); int bad = 0;
 81.     for (int k = 0; k < 20000; k++) {
 82.         uint64_t s = (((uint64_t)rand() << 31) ^ (uint64_t)rand()) & ((1ULL << 30) - 1);
 83.         int i = rand() % NG; uint32_t c = rand() % 8;
 84.         if (act(G[i], c, step(s)) != step(act(G[i], c, s))) bad++;
 85.     }
 86.     printf("  %s: %d linear maps commute with the partner; %d symmetries with translations;
commutation failures on 20,000 random states: %d\n",
 87.            USEP ? "native partner P" : "mirror J", NG, 8 * NG, bad);
 88.     uint64_t s0 = 1 << 3, s = s0; long long t = 0;
 89.     long long cnt[3] = {0, 0, 0}, kc[3][9]; memset(kc, 0, sizeof kc);
 90.     do {
 91.         s = step(s); t++;
 92.         uint32_t w = (s >> 3) & 7, D = (uint32_t)(s >> 6); int k = 0;
 93.         for (int m = 0; m < 8; m++) if ((D >> (3 * m)) & 7) k++;
 94.         cnt[carry_class(w)]++; kc[carry_class(w)][k]++;
 95.         if (D == 0 && s != s0) {
 96.             printf("  field at the mold at tick %lld: reader %s, carry %s;", t, nm(s & 7),
nm(w));
 97.             for (int i = 1; i < 8 * NG; i++) { if (act(G[i / 8], i % 8, s0) == s) { printf("
image of the start under g=(eR->%s, eL->%s, u->%s), translation %s;", nm(G[i / 8][0]), nm(G[i /
8][1]), nm(G[i / 8][2]), nm(i % 8)); break; } }
 98.             printf("\n");
 99.         }
100.         if (USEP && t == 4096) printf("  tick 4,096: reader %s, carry %s (carry class %s), kept
cells %d\n", nm(s & 7), nm(w), carry_class(w) == 0 ? "R1" : carry_class(w) == 1 ? "R2" : "R3", k);
101.     } while (s != s0);
102.     printf("  full period %lld\n", t);
```

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```
103.     if (USEP) {
104.         long long tot[9] = {0}; double mean = 0;
105.         for (int c = 0; c < 3; c++) for (int k = 0; k < 9; k++) { tot[k] += kc[c][k]; mean +=
(double)k * kc[c][k]; }
106.         printf("  kept cells k = 0..8 over the period:"); for (int k = 0; k < 9; k++) printf("
%lld", tot[k]); printf("\n");
107.         printf("  mean kept cells %.6f ; price of a move out of the carrier 2k = %.6f ; ticks
with k >= 7: %.4f%%\n", mean / t, 2 * mean / t, 100.0 * (tot[7] + tot[8]) / t);
108.         const char *L[3] = {"R1 (0, u)", "R2 (s, u+s)", "R3 (one break)"};
109.         for (int c = 0; c < 3; c++) { printf("  carry in %-15s %10lld ticks (%.4f%%) by k:",
L[c], cnt[c], 100.0 * cnt[c] / t); for (int k = 0; k < 9; k++) printf(" %lld", kc[c][k]);
printf("\n"); }
110.     }
111. }
112. static void births_reduced(void) {
113.     for (uint32_t w0 = 0; w0 < 8; w0++) {
114.         uint64_t s0 = (uint64_t)w0 << 3, s = s0; long long t = 0; double occ = 0;
115.         do { s = step(s); t++; uint32_t D = (uint32_t)(s >> 6); for (int m = 0; m < 8; m++) if
((D >> (3 * m)) & 7) occ += 1; } while (s != s0);
116.         printf("  carry %-5s period %-9lld mean kept cells %.6f\n", nm(w0), t, occ / t);
117.     }
118. }
119.
120. int main(void) {
121.     printf("PART A  every carry on an empty field, full word space\n");
122.     for (int w = 2; w <= 5; w++) partA(w);
123.     printf("\nPART B  the eight-word carrier, every even width >= 4\n");
124.     USEP = 1; births_reduced(); partB();
125.     printf("\n  mirror only\n");
126.     USEP = 0; births_reduced(); partB();
127.     return 0;
128. }
129.
130.
```

PART A every carry on an empty field, full word space
 n=2 carry 00 R1 period 6 field at the mold on 4 ticks first: t=2 p=00 w=11
 n=2 carry 01 R2 period 132 field at the mold on 1 ticks
 n=2 carry 10 R2 period 132 field at the mold on 1 ticks
 n=2 carry 11 R1 period 6 field at the mold on 4 ticks first: t=1 p=11 w=00
 n=2 summary: R1 2 carries (period 6), R2 2 carries (period 132), R3 0 carries (period 0)
 n=3 carry 000 R1 period 6 field at the mold on 4 ticks first: t=2 p=000 w=111
 n=3 carry 001 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=111 w=110
 n=3 carry 010 R2 period 132 field at the mold on 1 ticks
 n=3 carry 011 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=111 w=100
 n=3 carry 100 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=111 w=011
 n=3 carry 101 R2 period 132 field at the mold on 1 ticks
 n=3 carry 110 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=111 w=001
 n=3 carry 111 R1 period 6 field at the mold on 4 ticks first: t=1 p=111 w=000
 n=3 summary: R1 2 carries (period 6), R2 2 carries (period 132), R3 4 carries (period 43155322)
 n=4 carry 0000 R1 period 6 field at the mold on 4 ticks first: t=2 p=0000 w=1111
 n=4 carry 0001 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=1110
 n=4 carry 0010 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=1101
 n=4 carry 0011 R2 period 132 field at the mold on 1 ticks
 n=4 carry 0100 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=1011
 n=4 carry 0101 R2 period 132 field at the mold on 1 ticks

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n=4 carry 0110 R2 period 132 field at the mold on 1 ticks
n=4 carry 0111 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=1000
n=4 carry 1000 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=0111
n=4 carry 1001 R2 period 132 field at the mold on 1 ticks
n=4 carry 1010 R2 period 132 field at the mold on 1 ticks
n=4 carry 1011 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=0100
n=4 carry 1100 R2 period 132 field at the mold on 1 ticks
n=4 carry 1101 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=0010
n=4 carry 1110 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=1111 w=0001
n=4 carry 1111 R1 period 6 field at the mold on 4 ticks first: t=1 p=1111 w=0000
n=4 summary: R1 2 carries (period 6), R2 6 carries (period 132), R3 8 carries (period 43155322)
n=5 carry 00000 R1 period 6 field at the mold on 4 ticks first: t=2 p=00000 w=11111
n=5 carry 00001 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=11110
n=5 carry 00010 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=11101
n=5 carry 00011 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=11100
n=5 carry 00100 R2 period 132 field at the mold on 1 ticks
n=5 carry 00101 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=11010
n=5 carry 00110 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=11001
n=5 carry 00111 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=11000
n=5 carry 01000 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=10111
n=5 carry 01001 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=10110
n=5 carry 01010 R2 period 132 field at the mold on 1 ticks
n=5 carry 01011 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=10100
n=5 carry 01100 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=10011
n=5 carry 01101 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=10010
n=5 carry 01110 R2 period 132 field at the mold on 1 ticks
n=5 carry 01111 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=10000
n=5 carry 10000 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=01111
n=5 carry 10001 R2 period 132 field at the mold on 1 ticks
n=5 carry 10010 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=01101
n=5 carry 10011 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=01100
n=5 carry 10100 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=01011
n=5 carry 10101 R2 period 132 field at the mold on 1 ticks
n=5 carry 10110 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=01001
n=5 carry 10111 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=01000
n=5 carry 11000 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=00111
n=5 carry 11001 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=00110
n=5 carry 11010 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=00101
n=5 carry 11011 R2 period 132 field at the mold on 1 ticks
n=5 carry 11100 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=00011
n=5 carry 11101 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=00010
n=5 carry 11110 R3 period 43155322 field at the mold on 2 ticks first: t=21577661 p=11111 w=00001
n=5 carry 11111 R1 period 6 field at the mold on 4 ticks first: t=1 p=11111 w=00000
n=5 summary: R1 2 carries (period 6), R2 6 carries (period 132), R3 24 carries (period 43155322)

PART B the eight-word carrier, every even width >= 4
carry 0 period 6 mean kept cells 0.333333
carry eR period 43155322 mean kept cells 6.919906
carry eL period 43155322 mean kept cells 6.919906

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```
 carry s    period 132      mean kept cells 2.515152
 carry u    period 6        mean kept cells 0.333333
 carry u+eR  period 43155322  mean kept cells 6.919906
 carry u+eL  period 43155322  mean kept cells 6.919906
 carry u+s   period 132      mean kept cells 2.515152
 native partner P: 4 linear maps commute with the partner; 32 symmetries with translations; commutation
failures on 20,000 random states: 0
 tick 4,096: reader 0, carry u (carry class R1), kept cells 7
 field at the mold at tick 21577661: reader u, carry u+eR; image of the start under g=(eR->u+eR, eL->u+eL, u-
>u), translation u;
 full period 43155322
 kept cells k = 0..8 over the period: 2 106 3212 50576 475298 2738622 9409712 17402406 13075388
 mean kept cells 6.919906 ; price of a move out of the carrier 2k = 13.839812 ; ticks with k >= 7: 70.6235%
 carry in R1 (0, u)        9961342 ticks (23.0825%) by k: 0 8 342 7328 83790 552976 2079776 4077948 3159174
 carry in R2 (s, u+s)      10452516 ticks (24.2207%) by k: 0 12 552 10626 109006 649524 2263346 4219624
3199826
 carry in R3 (one break)    22741464 ticks (52.6968%) by k: 2 86 2318 32622 282502 1536122 5066590
9104834 6716388

 mirror only
 carry 0    period 1       mean kept cells 0.000000
 carry eR    period 518      mean kept cells 3.011583
 carry eL    period 518      mean kept cells 3.011583
 carry s    period 6       mean kept cells 1.166667
 carry u    period 6       mean kept cells 1.166667
 carry u+eR  period 518      mean kept cells 3.011583
 carry u+eL  period 518      mean kept cells 3.011583
 carry u+s   period 6       mean kept cells 1.166667
 mirror J: 8 linear maps commute with the partner; 64 symmetries with translations; commutation failures
on 20,000 random states: 0
 field at the mold at tick 259: reader 0, carry eL; image of the start under g=(eR->eL, eL->eR, u->u),
translation 0;
 full period 518

real    0m29.726s
user    0m29.444s
sys     0m0.000s
```

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```
"""Retained Continuation: the center of the native ring and the compiler identities.
Dean A. Kulik, QuHarmonics Research Group, ORCID 0009-0003-3128-8828. Runs by Claude (Anthropic), November 2026.
Run: python3 NEXUS_Retained_Continuation_center.py   (standard library only, a few seconds)"""
print("=== THE SEAM AND THE CENTER CARRIER ===")
# The center of the native ring: what sits at the seam, and where the center carrier lives.
def J(x,n): return int(format(x,'0%db'%n)[::-1],2)
def P(x,n): return x^((1<<n)-1) if J(x,n)==x else J(x,n)
def ring(n):
    reps=[x for x in range(1<<n) if x<P(x,n)]
    m=reps[-1]
    return reps[:-1]+[P(m,n),m]+[P(x,n) for x in reps[-2::-1]]
b=lambda x,n: format(x,'0%db'%n)
for n in range(2,15):
    R=ring(n); N=1<<n; h=N//2; READ={w:a for a,w in enumerate(R)}
    assert all(R[N-1-i]==P(R[i],n) for i in range(N))
    u=N-1
    line=f"n={n:2d} seam R[h-1],R[h] = {b(R[h-1],n)} {b(R[h],n)}  xor={b(R[h-1]^R[h],n)}"
    if n%2==0:
        r=n//2; eR=1<<(r-1); eL=1<<r
        assert (R[h-1],R[h])==(u^eR,u^eL)
        L={'0':0,'u':u,'eR':eR,'eL':eL,'s':eR^eL,'u+eR':u^eR,'u+eL':u^eL,'u+s':u^eR^eL}
        line+="  center carrier L at "+", ".join(f"{k}:{READ[v]}" for k,v in L.items())
        if n>=6:
            d=r-2; core=[0b0001,0b0011,0b0111,0b0101]
            ws=[(((1<<d)-1)<<(d+4))|(c<<d)|((1<<d)-1) for c in core]
            line+=f"  diamond-h={[READ[w]-h for w in ws]}"
    else:
        c=1<<((n-1)//2)
        line+=f"  u^c={b(u^c,n)} at {READ[u^c]}  c at {READ[c]}"
    print(line)
print("=== THE COMPILER ===")
# stored circle and its difference loop (bit k = s_k, MSB first; (T s)_k = s_(k+1))
R2=ring(2); s=''.join(format(w,'02b') for w in R2); print('R2',R2,'circle',s,hex(int(s,2)))
d=''.join(str(int(s[k])^int(s[(k+1)%8])) for k in range(8)); print('(I+T) circle',d,hex(int(d,2)))
sheet=''.join('1' if c=='0' else '0' for c in s[::-1]);
ds=''.join(str(int(sheet[k])^int(sheet[(k+1)%8])) for k in range(8))
print('sheet',sheet,hex(int(sheet,2)),'diff',ds,hex(int(ds,2)))
print('J8(circle)=',s[::-1],' T^5 circle=',s[5:]+s[:5])
q=''.join(str(int(s[k])^int(s[(k+1)%8])^int(s[(k+2)%8])) for k in range(8)); print('(I+T+T^2) circle',q,hex(int(q,2)))
# full adder truth tables (index abc = 000..111, MSB = index 0)
print('sum',''.join(str(a^b^c) for a in (0,1) for b in (0,1) for c in (0,1)),'carry',''.join(str(int(a+b+c>=2)) for a in (0,1) for b in (0,1) for c in (0,1)))
# triad Gamma on a ring: (a,v)->(READ v, v ^ R[a]) ; period-3 check and class counts
for n in range(1,7):
    R=ring(n) if n>1 else [0,1]; READ={w:a for a,w in enumerate(R)}; N=1<<n
    G=lambda a,v:(READ[v],v^R[a]); ok=all(G(*G(*G(a,v)))==(a,v) for a in range(N) for v in range(N))
    zero=sum(1 for a in range(N) for v in range(N) if v==R[a] and v!=0)  # zero-gap states
    print(f'n={n} Gamma^3=I {ok}  4^n={4**n} = 1 + 3*{(4**n-1)//3}  zero-gap triads {2**n-1} interior triads {(2**n-1)*(2**n-2)//3}  lines L_n={(2**n-1)*(2**n-2)//6}')
# route-switch defect zeros and perimeter lines
for n in range(2,11):
    R=ring(n); N=1<<n
    t=[R[a]^R[(a+1)%N]^R[(a+2)%N] for a in range(N)]
    zeros=[a for a in range(N) if t[a]==0]
    lines=set()
    for a in range(N):
        x,y=R[a],R[(a+1)%N]
        if x and y and x!=y: lines.add(frozenset((x,y,x^y)))
```

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```
56.     print(f'n={n} t_a zeros at {zeros}  distinct projective lines on perimeter edges
{len(lines)}')
57.
```

```
=== THE SEAM AND THE CENTER CARRIER ===
n= 2 seam R[h-1],R[h] = 10 01  xor=11  center carrier L at 0:0, U:3, eR:2, eL:1, s:3, U+eR:1, U+eL:2, U+s:0
n= 3 seam R[h-1],R[h] = 110 011  xor=101  U^c=101 at 5  c at 2
n= 4 seam R[h-1],R[h] = 1101 1011  xor=0110  center carrier L at 0:0, U:15, eR:2, eL:13, s:5, U+eR:7, U+eL:8,
U+s:10
n= 5 seam R[h-1],R[h] = 11101 10111  xor=01010  U^c=11011 at 27  c at 4
n= 6 seam R[h-1],R[h] = 111011 110111  xor=001100  center carrier L at 0:0, U:63, eR:4, eL:59, s:11, U+eR:31,
U+eL:32, U+s:52  diamond-h=[-6, -4, -2, -3]
n= 7 seam R[h-1],R[h] = 1111011 1101111  xor=0010100  U^c=1110111 at 119  c at 8
n= 8 seam R[h-1],R[h] = 11110111 11101111  xor=00011000  center carrier L at 0:0, U:255, eR:8, eL:247, s:23,
U+eR:127, U+eL:128, U+s:232  diamond-h=[-6, -4, -2, -3]
n= 9 seam R[h-1],R[h] = 111110111 111011111  xor=000101000  U^c=111101111 at 495  c at 16
n=10 seam R[h-1],R[h] = 1111101111 1111011111  xor=0000110000  center carrier L at 0:0, U:1023, eR:16,
eL:1007, s:47, U+eR:511, U+eL:512, U+s:976  diamond-h=[-6, -4, -2, -3]
n=11 seam R[h-1],R[h] = 11111101111 11110111111  xor=00001010000  U^c=11111011111 at 2015  c at 32
n=12 seam R[h-1],R[h] = 111111011111 111110111111  xor=000001100000  center carrier L at 0:0, U:4095,
eR:32, eL:4063, s:95, U+eR:2047, U+eL:2048, U+s:4000  diamond-h=[-6, -4, -2, -3]
n=13 seam R[h-1],R[h] = 1111111011111 1111101111111  xor=0000010100000  U^c=1111110111111 at 8127  c at
64
n=14 seam R[h-1],R[h] = 11111110111111 11111101111111  xor=00000011000000  center carrier L at 0:0,
U:16383, eR:64, eL:16319, s:191, U+eR:8191, U+eL:8192, U+s:16192  diamond-h=[-6, -4, -2, -3]
=== THE COMPILER ===
R2 [0, 2, 1, 3] circle 00100111 0x27
(I+T) circle 01101001 0x69
sheet 00011011 0x1b diff 00101101 0x2d
J8(circle)= 11100100  T^5 circle= 11100100
(I+T+T^2) circle 11110101 0xf5
sum 01101001 carry 00010111
n=1 Gamma^3=I True  4^n=4 = 1 + 3*1  zero-gap triads 1  interior triads 0  lines L_n=0
n=2 Gamma^3=I True  4^n=16 = 1 + 3*5  zero-gap triads 3  interior triads 2  lines L_n=1
n=3 Gamma^3=I True  4^n=64 = 1 + 3*21  zero-gap triads 7  interior triads 14  lines L_n=7
n=4 Gamma^3=I True  4^n=256 = 1 + 3*85  zero-gap triads 15  interior triads 70  lines L_n=35
n=5 Gamma^3=I True  4^n=1024 = 1 + 3*341  zero-gap triads 31  interior triads 310  lines L_n=155
n=6 Gamma^3=I True  4^n=4096 = 1 + 3*1365  zero-gap triads 63  interior triads 1302  lines L_n=651
n=2 t_a zeros at [1]  distinct projective lines on perimeter edges 1
n=3 t_a zeros at [3]  distinct projective lines on perimeter edges 5
n=4 t_a zeros at [1, 3, 12]  distinct projective lines on perimeter edges 11
n=5 t_a zeros at [1, 28]  distinct projective lines on perimeter edges 28
n=6 t_a zeros at [1, 60]  distinct projective lines on perimeter edges 60
n=7 t_a zeros at [1, 124]  distinct projective lines on perimeter edges 124
n=8 t_a zeros at [1, 252]  distinct projective lines on perimeter edges 252
n=9 t_a zeros at [1, 508]  distinct projective lines on perimeter edges 508
n=10 t_a zeros at [1, 1020]  distinct projective lines on perimeter edges 1020
```

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